Gibran Akmal

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EDUCATION

University of Calgary **Calgary, AB**
Bachelor of Science, in Software Engineering *Graduated with 3.6 GPA*

WORK EXPERIENCE

Critical Control Energy Services **Calgary, AB**
Data/Business Analyst *Oct 2025 – Present*

- Facilitate the translation of complex client business needs into technical specifications, managing **20+ Product Backlog Items (PBIs)** per sprint within Azure DevOps (**ADO**) to align developer workflows with system architecture requirements.
- Act as the technical lead for a **4-person development team**, streamlining the delivery of system enhancements by defining clear acceptance criteria, resulting in a **20%** reduction in rework during the User Acceptance Testing (UAT) phase.
- Built and deployed a **Claude LLM** pipeline triggered via FrontApp web-hooks to automate client email triage, using prompt engineering to classify request intent across 5 ProTrend database change categories (e.g. cost centre updates, asset changes) and trigger validated API **POST** requests, cutting manual processing time by ~**70%** across ~**80** monthly requests with human-in-the-loop review enforcing AER data integrity. Further refined this workflow to generate and send automated email replies/confirmations to clients for changes made.
- Automated validation checks (using **VBA**) for Electronic Flow Measurement (EFM) and liquid calibration reports, implementing QA checks to ensure compliance with AER/Directive 017 standards and reducing manual audit time by **25%.**
- Engineered and maintained automated **Python/SQL** scripts to extract measurement data from external vendors (Cenozon, CEG, CFI) and transform it into proprietary ProTrend schemas, automating the ingestion of **15,000+ monthly data points** into production databases.

ExxonMobil **Calgary, AB**
Data Engineer *Jan 2025 – Oct 2025*

- Engineered a data pipeline for near real time streaming of Caterpillar's autonomous haul truck telemetry data (Minestar) via **Azure IoT Hub** and **MQTT** into Delta tables, enabling **0.5 km/h** increase in average truck speed and **$11M** in operational savings.
- Maintained **10+ CI/CD** pipelines for deployment workflows across **3** environments using **GitHub Actions** and implemented **Terraform** for automated infrastructure provisioning reducing release errors by **18%**.
- Built an **Azure Data Factory** Pipeline to automate the ingestion and validation of **EMAT** and **UT** laser scan files at the Kearl asset for Large Diameter Piping, increasing data availability for analytics by **32%** and reducing manual processing times by **20+** hours/week.
- Optimized spark jobs and tailored **Databricks** compute clusters to cut data ingestion times by **20%** for mobile compatible site critical applications such as FieldHub and MaintenanceHub which is utilized by over **2000** field technicians.
- Collaborated with data science teams to construct a predictive analytics tool (using **pandas** and **scikit-learn**) to forecast wear on critical spools and automatically suggest degrees of rotation to maximize spool life, preventing early failures and enabling **$13M** in cost avoidance via proactive inspection planning.
- Acted as a liaison between engineering and business teams during pipeline tool development, translating business needs into technical requirements and technical constraints back to stakeholders for **30+ sensitive spools**, ensuring alignment that led to successful deployment and adoption across three operational units.

TC Energy **Calgary, AB**
Pipeline Integrity Engineering Intern *May 2023 – Aug 2024*

- Developed a statistical risk tool using **Python** libraries to forecast leak probability from wall thickness data, enabling pipeline integrity teams to reprioritize inspections and cut risk exposure by **15%.** I further utilized this tool for validating Safe Excavation Pressure recommendations when drafting SEP Tech Memos for **20+** critical lines.
- Translated findings from **15+** field site visits along with engineering deliverables such as plot plans, scope sheets, tech memos, as-builts and single line diagrams into assessment briefs for senior engineering managers, directly informing **$2M+** in pipeline maintenance prioritization.
- Developed and trained and object detection model through leveraging the **PyTorch** machine learning framework to automatically detect and classify corrosion on pipeline surfaces from inspection images. This model achieved a **mean average precision** of **76%**
- Collaborated with the valve integrity team to validate **5+** actuator designs proposed for ball and gate valves for a line upgrade to improve gas flow by **12%** per outage cycle
- Delivered a **Tableau** dashboard integrating **SQL** datasets via **Denodo** virtual views to correlate ILI-detected metal loss, SCC, and dent features with field-verified integrity defects, improving anomaly validation accuracy by **20%**.

PROJECT EXPERIENCE

Race-Car Telemetry System – Awarded 2nd place at the Schulich Engineering Design Fair **Calgary, AB**
Hardware and Firmware Team Member – Schulich Formula Racing (SAE Electric) *Sep 2024 - April 2025*

- Designed and implemented a firmware solution for an **ESP-32** microcontroller to transmit CAN packets over Wi-Fi to a custom **Flask**-based web application, **eliminating** the need for post-race physical data dumps and enabling **live** vehicle diagnostics for competition.
- Delivered a structured business case for my solution to over **20** industry judges—including engineers and executives from companies

like Garmin, Shell, and Nvidia—demonstrating real-world viability and scalability for motorsport and fleet monitoring applications.

- Conducted market validation with **12+** race engineers and student motorsport teams, aligning product features with real-world needs and achieving a **90%** positive feedback rate based on user-fit prior to the initial prototype.
- Coordinated development timelines across a **4-**person hardware team, breaking down technical milestones into weekly deliverables and tracking progress with Gantt charts, which ensured on-time delivery of the system ahead of the final design review by **10** days.
- Outlined a go-to-market plan targeting university race teams and amateur leagues, projecting initial deployment in **10+** teams with **<$250** hardware cost, and validated interest through post-event surveys with **80%** positive intent.

LEADERSHIP EXPERIENCE

Step Forward Volunteering Association **Calgary, AB**

Member *Feb 2020 – June 2025*

- I assisted in the conception and execution of a charitable walkathon in support of the Students Against Domestic Abuse Association (SADAA) to raise awareness and funds for their cause of combating domestic abuse, and we managed to raise about **$20,000.**
- Organized food drives through **12+** donation stations set up across the University of Calgary to distribute sustenance to those in need.
- Contributed to the setup of the "Disney on Ice" community event by running the toy assembly station and serving children at snack shops.

SKILLS

.NET, C++, C#, Java, Go, JavaScript, SQL, React.js, MLOps, NLP, Kubernetes, Docker, Git, Linux, OpenCV, MATLAB, TensorFlow, PowerBi